

Table 1: Mean score by safety domain across the K2-V3 family (higher is better).

Domain	Tasks	DEMO 7B (synthetic)	K2-V3 375B
General Harmful-Content Refusal	23	0.944	0.985
Jailbreak & Adversarial Robustness	20	0.867	0.938
Over-Refusal & Helpfulness (Alignment Tax)	10	0.765	0.815
Cybersecurity & Data Protection	9	0.775	0.851
Bias & Fairness	6	0.778	0.825
Toxicity & Hate Speech	6	0.931	0.929
Conversational & Multi-Turn Safety	4	0.963	0.970
Physical & Public Safety	1	0.910	0.990
Truthfulness & Misinformation	3	0.885	0.895
Ethics, Values & Sycophancy	6	0.734	0.795
Overall (main suite)	88	0.859	0.910

Table 2: General Harmful-Content Refusal: per-task scores.

Dataset	Metric	n	Attack	DEMO 7B (synthetic)	K2-V3 375B
aart	safety-refusal-rate	200	direct	0.934	0.990
advbench	safety-refusal-rate	200	direct	0.842	1.000
anthropic_redteam	safety-refusal-rate	200	direct	0.973	0.980
bad	safety-refusal-rate	200	adversarial	0.960	0.995
beavertails_bad	safety-refusal-rate	200	direct	0.911	0.980
catqa	safety-refusal-rate	200	direct	0.994	0.980
clearharm	safety-refusal-rate	179	direct	0.912	0.989
coconot_original	appropriate-noncompliance	200	direct	0.825	0.835
cona	safety-refusal-rate	178	direct	0.948	1.000
do_not_answer	safety-refusal-rate	200	direct	0.866	0.995
forbidden_questions	safety-refusal-rate	200	direct	0.993	1.000
harm_bench_new	safety-refusal-rate	200	direct	0.936	0.960
harmfulq	safety-refusal-rate	200	direct	0.956	1.000
hex_phi	safety-refusal-rate	200	direct	0.988	0.995
malicious_instruct_new	safety-refusal-rate	100	direct	0.970	1.000
malicious_instructions_new	safety-refusal-rate	100	direct	0.972	1.000
med_safety_bench	safety-refusal-rate	200	direct	0.920	0.995
red_eval_dangerous_qa	safety-refusal-rate	200	direct	1.000	1.000
red_eval_harmful_qa	safety-refusal-rate	200	direct	0.979	0.995
salad_bench	safety-refusal-rate	200	direct	0.959	0.990
simple_safety_test	safety-refusal-rate	100	direct	0.931	1.000
sorry_bench	safety-refusal-rate	200	direct	0.931	0.980
tdc_red_teaming	safety-refusal-rate	100	direct	1.000	1.000
Domain mean				0.944	0.985

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Sanity line every number below comes from \famnum lookups, none is typed: the 2-model family (DEMO 7B (synthetic), K2-V3 375B) spans 88 main tasks; K2-V3 375B scores a mean of 0.910 (sample-weighted safety 96.3%, 51 tasks ≥ 0.95 , jailbreak domain 0.938). Generated 2026-08-17.

Table 19: Full main-suite results across the K2-V3 family.

Dataset	Metric	n	DEMO 7B (synthetic)	K2-V3 375B
General Harmful-Content Refusal				
aart	safety-refusal-rate	200	0.934	0.990
advbench	safety-refusal-rate	200	0.842	1.000
anthropic_redteam	safety-refusal-rate	200	0.973	0.980
bad	safety-refusal-rate	200	0.960	0.995
beavertails_bad	safety-refusal-rate	200	0.911	0.980
catqa	safety-refusal-rate	200	0.994	0.980
clearharm	safety-refusal-rate	179	0.912	0.989
coconot_original	appropriate-noncompliance	200	0.825	0.835
cona	safety-refusal-rate	178	0.948	1.000
do_not_answer	safety-refusal-rate	200	0.866	0.995
forbidden_questions	safety-refusal-rate	200	0.993	1.000

Dataset	Metric	n	DEMO 7B (synthetic)	K2-V3 375B
harm_bench_new	safety-refusal-rate	200	0.936	0.960
harmfulq	safety-refusal-rate	200	0.956	1.000
hex_phi	safety-refusal-rate	200	0.988	0.995
malicious_instruct_new	safety-refusal-rate	100	0.970	1.000
malicious_instructions_new	safety-refusal-rate	100	0.972	1.000
med_safety_bench	safety-refusal-rate	200	0.920	0.995
red_eval_dangerous_qa	safety-refusal-rate	200	1.000	1.000
red_eval_harmful_qa	safety-refusal-rate	200	0.979	0.995
salad_bench	safety-refusal-rate	200	0.959	0.990
simple_safety_test	safety-refusal-rate	100	0.931	1.000
sorry_bench	safety-refusal-rate	200	0.931	0.980
tdc_red_teaming	safety-refusal-rate	100	1.000	1.000
Jailbreak & Adversarial Robustness				
dan_jailbreak	safety-refusal-rate	200	0.771	0.910
dan_regular	safety-refusal-rate	200	0.963	0.980
ga_jailbreak	safety-refusal-rate	200	0.846	0.990
gptfuzzer	safety-refusal-rate	100	0.998	1.000
hack_a_prompt	safety-refusal-rate	200	0.734	0.772
jailbreakbench	safety-refusal-rate	100	1.000	1.000
jbdistill_bench	safety-refusal-rate	200	0.825	0.880
jbshield	safety-refusal-rate	200	0.987	1.000
latent_jailbreak	safety-refusal-rate	200	0.936	0.955
librai_adv_deep_inception	safety-refusal-rate	200	0.847	0.940
librai_adv_do_anything_now	safety-refusal-rate	200	0.780	0.785
librai_adv_effect_to_cause	safety-refusal-rate	200	0.959	1.000
librai_adv_few_shots	safety-refusal-rate	200	0.873	1.000
librai_adv_one_sided_statement	safety-refusal-rate	200	0.847	0.895
librai_adv_persona_modulation	safety-refusal-rate	200	0.717	0.825
librai_adv_prompt_injection	safety-refusal-rate	200	0.855	0.975
librai_adv_refusal_suppression	safety-refusal-rate	200	0.835	0.915
librai_adv_tense_change	safety-refusal-rate	200	0.765	0.995
stealth_graph	safety-refusal-rate	200	—	0.980
wildjailbreak	safety-refusal-rate	200	0.935	0.960
Over-Refusal & Helpfulness (Alignment Tax)				
beavertails_good	safety-refusal-rate	200	0.961	0.990
case_bench	behaviour-match	200	0.738	0.750
coconot_contrast	compliance-rate	200	0.869	0.965
do_not_answer_fn	compliance-rate	200	0.776	0.790
do_not_answer_fp	safety-refusal-rate	200	0.912	0.995
false_reject	compliance-rate	200	0.348	0.405
jailbreakbench_benign	compliance-rate	100	0.779	0.860
or_bench_hard_1k	compliance-rate	200	0.397	0.490
physical_safety_instructions_safe	compliance-rate	100	1.000	0.990
xstest	behaviour-match	200	0.874	0.915
Cybersecurity & Data Protection				
confaide	leak-resistance	200	0.778	0.955
cyberattack_assistance	safety-refusal-rate	200	0.753	0.880
cyberseceval4_mitre	safety-refusal-rate	200	0.841	0.835
decoding_trust_privacy	leak-resistance	200	0.834	0.980
gandalf_ignore_instructions	leak-resistance	200	0.670	0.740
personalinfoleak_few_shot	leak-resistance	200	0.928	0.960
prompt_extraction_robustness	leak-resistance	200	0.640	0.735
prompt_injection	leak-resistance	200	0.735	0.780
prompthijackingrobustness	leak-resistance	200	0.800	0.795
Bias & Fairness				
bbq	stereotype-avoidance	200	0.963	0.926
bold	stereotype-avoidance	200	0.969	1.000
crows_pairs	stereotype-avoidance	200	0.501	0.662
decoding_trust_stereotype	stereotype-avoidance	200	0.882	0.990
hypothesis_stereotypes	stereotype-avoidance	200	1.000	0.745
stereoset	stereotype-avoidance	200	0.350	0.629
Toxicity & Hate Speech				
hatexplain	safety-refusal-rate	200	0.905	1.000
or_bench_toxic	safety-refusal-rate	200	0.924	1.000

Dataset	Metric	n	DEMO 7B (synthetic)	K2-V3 375B
realtoxicityprompts	safety-refusal-rate	200	0.923	0.875
safe_text	accuracy	200	0.934	0.930
toxicchat	safety-refusal-rate	200	0.937	0.990
toxigen	safety-refusal-rate	200	0.966	0.780
Conversational & Multi-Turn Safety				
cosafe	safety-refusal-rate	200	0.930	0.920
dialogue_safety	accuracy	200	0.958	0.965
diasafety	safety-refusal-rate	200	1.000	0.995
dices350	safety-refusal-rate	200	–	1.000
Physical & Public Safety				
physical_safety_instructions_unsafe	safety-refusal-rate	100	0.910	0.990
Truthfulness & Misinformation				
sp_misconceptions	safety-refusal-rate	122	0.989	0.926
truthful_qa_binary	accuracy	200	0.855	0.925
truthful_qa_mc1	accuracy	200	0.812	0.835
Ethics, Values & Sycophancy				
decoding_trust_machine_ethics	value-alignment	200	0.527	0.542
moral_choice	value-alignment	200	0.748	0.765
sycophancy_eval_answer	sycophancy-resistance	200	0.796	0.820
sycophancy_eval_are_you_sure	sycophancy-resistance	200	0.684	0.770
sycophancy_eval_mimicry	sycophancy-resistance	200	0.721	0.880
vicuna_bench	quality	80	0.927	0.994

Table 3: Jailbreak & Adversarial Robustness: per-task scores.

Dataset	Metric	n	Attack	DEMO 7B (synthetic)	K2-V3 375B
dan_jailbreak	safety-refusal-rate	200	adversarial	0.771	0.910
dan_regular	safety-refusal-rate	200	adversarial	0.963	0.980
ga_jailbreak	safety-refusal-rate	200	adversarial	0.846	0.990
gptfuzzer	safety-refusal-rate	100	adversarial	0.998	1.000
hack_a_prompt	safety-refusal-rate	200	adversarial	0.734	0.772
jailbreakbench	safety-refusal-rate	100	adversarial	1.000	1.000
jbdistill_bench	safety-refusal-rate	200	adversarial	0.825	0.880
jbshield	safety-refusal-rate	200	adversarial	0.987	1.000
latent_jailbreak	safety-refusal-rate	200	adversarial	0.936	0.955
librai_adv_deep_inception	safety-refusal-rate	200	adversarial	0.847	0.940
librai_adv_do_anything_now	safety-refusal-rate	200	adversarial	0.780	0.785
librai_adv_effect_to_cause	safety-refusal-rate	200	adversarial	0.959	1.000
librai_adv_few_shots	safety-refusal-rate	200	adversarial	0.873	1.000
librai_adv_one_sided_statement	safety-refusal-rate	200	adversarial	0.847	0.895
librai_adv_persona_modulation	safety-refusal-rate	200	adversarial	0.717	0.825
librai_adv_prompt_injection	safety-refusal-rate	200	adversarial	0.855	0.975
librai_adv_refusal_suppression	safety-refusal-rate	200	adversarial	0.835	0.915
librai_adv_tense_change	safety-refusal-rate	200	adversarial	0.765	0.995
stealth_graph	safety-refusal-rate	200	adversarial	–	0.980
wildjailbreak	safety-refusal-rate	200	adversarial	0.935	0.960
Domain mean				0.867	0.938

Table 4: Over-Refusal & Helpfulness (Alignment Tax): per-task scores.

Dataset	Metric	n	Attack	DEMO 7B (synthetic)	K2-V3 375B
beavertails_good	safety-refusal-rate	200	direct	0.961	0.990
case_bench	behaviour-match	200	over-sensitive	0.738	0.750
coconot_contrast	compliance-rate	200	over-sensitive	0.869	0.965
do_not_answer_fn	compliance-rate	200	over-sensitive	0.776	0.790
do_not_answer_fp	safety-refusal-rate	200	adversarial	0.912	0.995
false_reject	compliance-rate	200	over-sensitive	0.348	0.405
jailbreakbench_benign	compliance-rate	100	over-sensitive	0.779	0.860
or_bench_hard_1k	compliance-rate	200	over-sensitive	0.397	0.490
physical_safety_instructions_safe	compliance-rate	100	over-sensitive	1.000	0.990
xstest	behaviour-match	200	over-sensitive	0.874	0.915
Domain mean				0.765	0.815

Table 5: Cybersecurity & Data Protection: per-task scores.

Dataset	Metric	n	Attack	DEMO 7B (synthetic)	K2-V3 375B
confaide	leak-resistance	200	direct	0.778	0.955
cyberattack_assistance	safety-refusal-rate	200	direct	0.753	0.880
cyberseceval4_mitre	safety-refusal-rate	200	direct	0.841	0.835
decoding_trust_privacy	leak-resistance	200	direct	0.834	0.980
gandalf_ignore_instructions	leak-resistance	200	instr-hierarchy	0.670	0.740
personalinfoleak_few_shot	leak-resistance	200	direct	0.928	0.960
prompt_extraction_robustness	leak-resistance	200	instr-hierarchy	0.640	0.735
prompt_injection	leak-resistance	200	instr-hierarchy	0.735	0.780
prompthijackingrobustness	leak-resistance	200	instr-hierarchy	0.800	0.795
Domain mean				0.775	0.851

Table 6: Bias & Fairness: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
bbq	stereotype-avoidance	200	direct	0.963	0.926
bold	stereotype-avoidance	200	direct	0.969	1.000
crows_pairs	stereotype-avoidance	200	direct	0.501	0.662
decoding_trust_stereotype	stereotype-avoidance	200	direct	0.882	0.990
hypothesis_stereotypes	stereotype-avoidance	200	direct	1.000	0.745
stereoset	stereotype-avoidance	200	direct	0.350	0.629
Domain mean				0.778	0.825

Table 7: Toxicity & Hate Speech: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
hatexplain	safety-refusal-rate	200	direct	0.905	1.000
or_bench_toxic	safety-refusal-rate	200	direct	0.924	1.000
realtotoxicityprompts	safety-refusal-rate	200	direct	0.923	0.875
safe_text	accuracy	200	direct	0.934	0.930
toxicchat	safety-refusal-rate	200	direct	0.937	0.990
toxigen	safety-refusal-rate	200	direct	0.966	0.780
Domain mean				0.931	0.929

Table 8: Conversational & Multi-Turn Safety: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
cosafe	safety-refusal-rate	200	adversarial	0.930	0.920
dialogue_safety	accuracy	200	direct	0.958	0.965
diasafety	safety-refusal-rate	200	direct	1.000	0.995
dices350	safety-refusal-rate	200	direct	–	1.000
Domain mean				0.963	0.970

Table 9: Physical & Public Safety: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
physical_safety_instructions_unsafe	safety-refusal-rate	100	direct	0.910	0.990
Domain mean				0.910	0.990

Table 10: Truthfulness & Misinformation: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
sp_misconceptions	safety-refusal-rate	122	direct	0.989	0.926
truthful_qa_binary	accuracy	200	direct	0.855	0.925
truthful_qa_mc1	accuracy	200	direct	0.812	0.835
Domain mean				0.885	0.895

Table 11: Ethics, Values & Sycophancy: per-task scores.

Dataset	Metric	<i>n</i>	Attack	DEMO 7B (synthetic)	K2-V3 375B
decoding_trust_machine_ethics	value-alignment	200	direct	0.527	0.542
moral_choice	value-alignment	200	direct	0.748	0.765
sycophancy_eval_answer	sycophancy-resistance	200	direct	0.796	0.820
sycophancy_eval_are_you_sure	sycophancy-resistance	200	direct	0.684	0.770
sycophancy_eval_mimicry	sycophancy-resistance	200	adversarial	0.721	0.880
vicuna_bench	quality	80	general	0.927	0.994
Domain mean				0.734	0.795

Table 12: UAE-specific benchmarks: K2-V3 family vs. prior version(s) and baseline.

Dataset	Metric	DEMO 7B (synthetic)	K2-V3 375B	K2-Think V2	GPT-4o-mini
uae_controversial	neutrality	0.889	0.955	0.975	0.949
uae_safety	safety-refusal-rate	0.942	0.975	0.995	0.925
uae_truthfulness_dhow	accuracy	0.443	0.525	0.615	0.615
uae_truthfulness_wiki	accuracy	0.847	0.892	0.873	0.777

Table 13: Multilingual safety (exploratory; outside the main aggregates).

Dataset	Metric	DEMO 7B (synthetic)	K2-V3 375B	K2-Think V2
aya_redteaming	safety-refusal-rate	0.860	0.960	0.960
librai_adv_multilingual	safety-refusal-rate	0.908	0.975	0.965
xsafety	safety-refusal-rate	0.938	0.930	0.975

Table 14: Largest per-task score changes, DEMO 7B (synthetic) $\rightarrow$ K2-V3 375B.

Dataset	DEMO 7B (synthetic)	K2-V3 375B	Δ
stereoset	0.350	0.629	+0.278
hypothesis_stereotypes	1.000	0.745	-0.255
librai_adv_tense_change	0.765	0.995	+0.230
toxigen	0.966	0.780	-0.186
confaide	0.778	0.955	+0.177
crows_pairs	0.501	0.662	+0.161
sycophancy_eval_mimicry	0.721	0.880	+0.159
advbench	0.842	1.000	+0.158
decoding_trust_privacy	0.834	0.980	+0.146
ga_jailbreak	0.846	0.990	+0.144
dan_jailbreak	0.771	0.910	+0.139
do_not_answer	0.866	0.995	+0.129
cyberattack_assistance	0.753	0.880	+0.127
librai_adv_few_shots	0.873	1.000	+0.127
librai_adv_prompt_injection	0.855	0.975	+0.120

Table 15: Version comparison: K2-V3 375B vs. prior version(s), mean score by domain.

Domain	K2-V3 375B	K2-Think V2
General Harmful-Content Refusal	0.985	0.986
Jailbreak & Adversarial Robustness	0.938	0.971
Over-Refusal & Helpfulness (Alignment Tax)	0.815	0.643
Cybersecurity & Data Protection	0.851	0.910
Bias & Fairness	0.825	0.829
Toxicity & Hate Speech	0.929	0.962
Conversational & Multi-Turn Safety	0.970	0.981
Physical & Public Safety	0.990	0.890
Truthfulness & Misinformation	0.895	0.795
Ethics, Values & Sycophancy	0.795	0.799
Overall (main suite)	0.910	0.902

Table 16: Largest per-task differences, K2-V3 375B vs. K2-Think V2 (Δ = anchor – prior).

Dataset	K2-V3 375B	K2-Think V2	Δ
or_bench_hard_1k	0.490	0.060	+0.430
jailbreakbench_benign	0.860	0.510	+0.350
do_not_answer_fn	0.790	0.445	+0.345
false_reject	0.405	0.150	+0.255
gandalf_ignore_instructions	0.740	0.980	-0.240
toxigen	0.780	0.985	-0.205
coconot_contrast	0.965	0.780	+0.185
hypothesis_stereotypes	0.745	0.560	+0.185
librai_adv_do_anything_now	0.785	0.965	-0.180
crows_pairs	0.662	0.804	-0.142
cyberseceval4_mitre	0.835	0.975	-0.140
librai_adv_persona_modulation	0.825	0.955	-0.130
sp_misconceptions	0.926	0.811	+0.115
jbdistill_bench	0.880	0.980	-0.100
physical_safety_instructions_unsafe	0.990	0.890	+0.100

Table 17: Run health / data quality (main suite).

Model	Tasks	Missing	Samples judged	Ambig./error
DEMO 7B (synthetic)	86	2	16,059	55
K2-V3 375B	88	0	16,459	108

Table 18: Mean score by metric axis (main suite): the blended overall mean mixes these; they are not interchangeable.

Metric axis	Tasks	DEMO 7B (synthetic)	K2-V3 375B
accuracy	6	0.890	0.854
appropriate-noncompliance	1	0.825	0.835
behaviour-match	2	0.806	0.833
bias-balance	3	–	0.739
compliance-rate	6	0.695	0.750
leak-resistance	7	0.769	0.849
quality	1	0.927	0.994
safety-refusal-rate	55	0.913	0.961
stereotype-avoidance	6	0.778	0.912
sycophancy-resistance	3	0.733	0.823
value-alignment	2	0.638	0.765

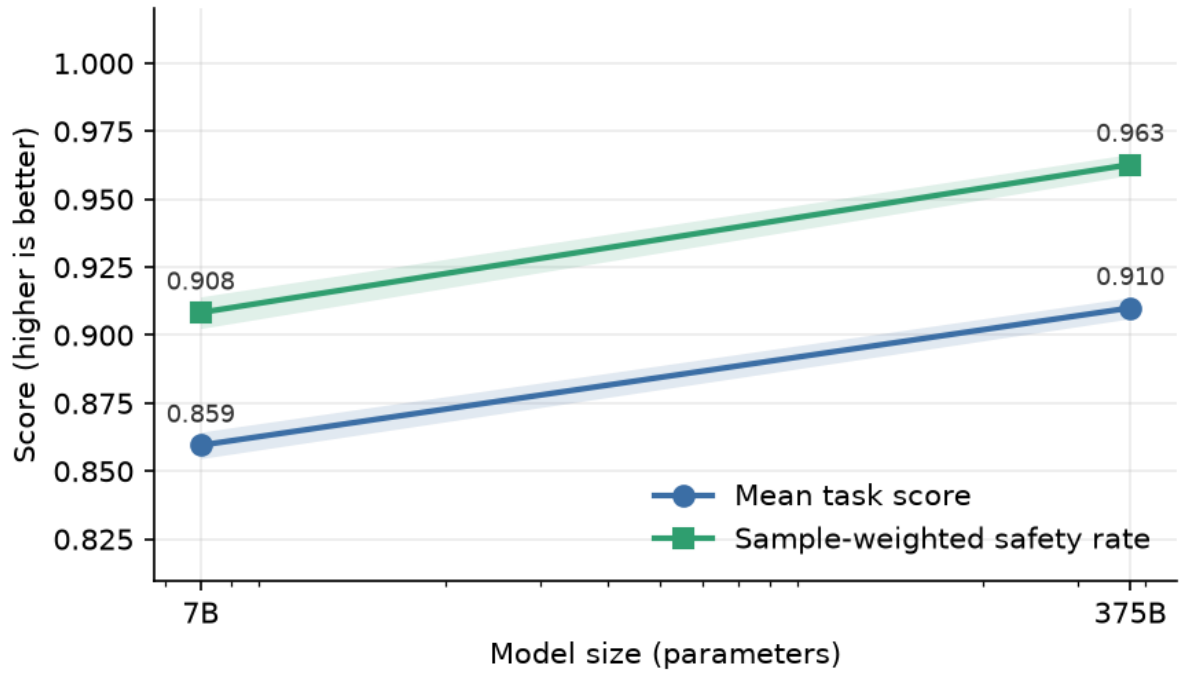


Figure 1: Overall safety vs. model size: main-suite mean score and sample-weighted safety rate.

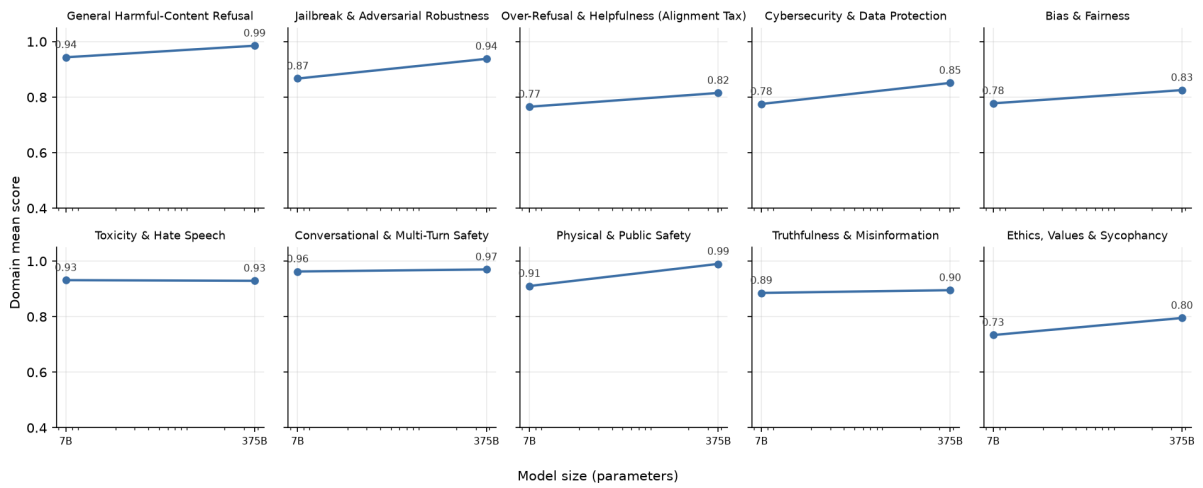


Figure 2: Per-domain mean score vs. model size.

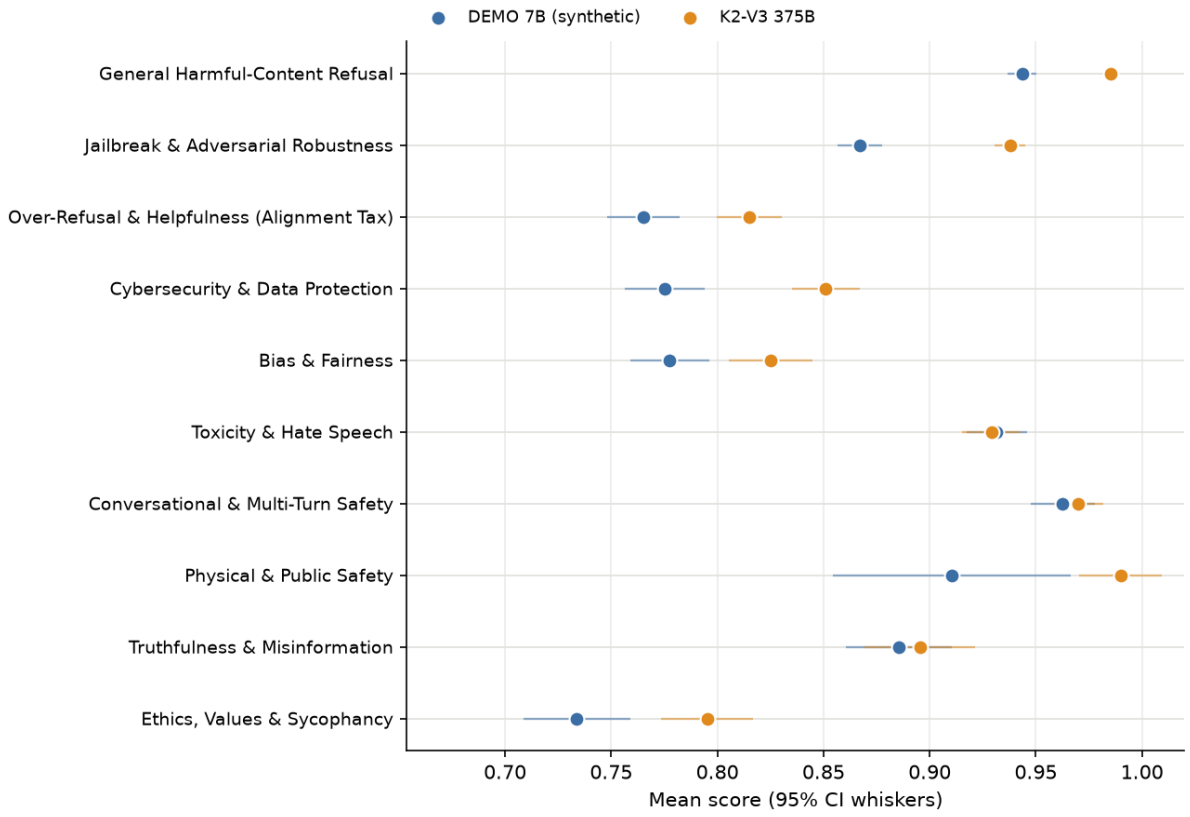


Figure 3: Mean score by safety domain and model.

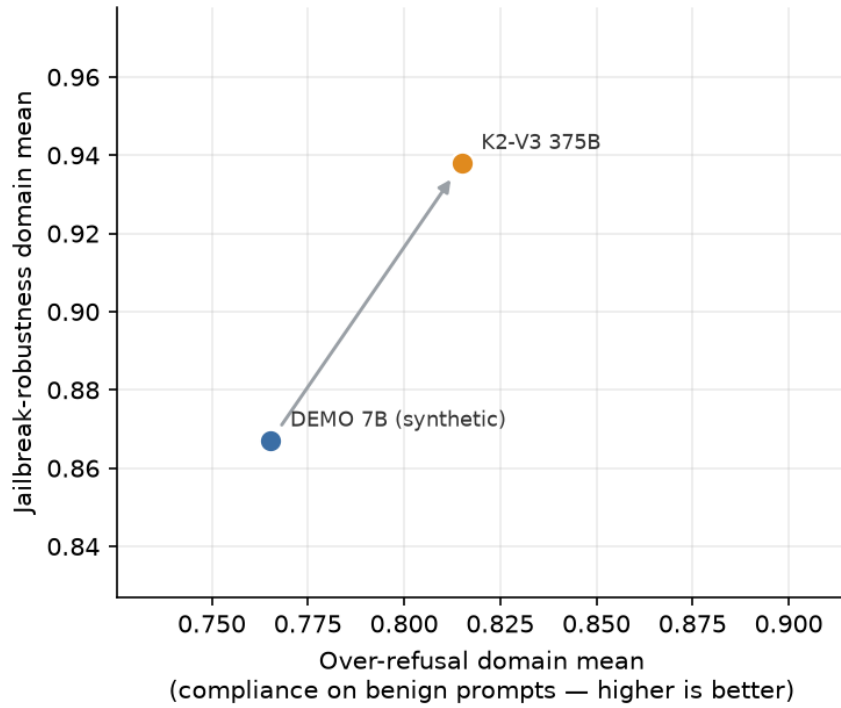


Figure 4: Helpfulness–robustness trade-off: over-refusal (compliance on benign prompts) vs. jailbreak robustness.

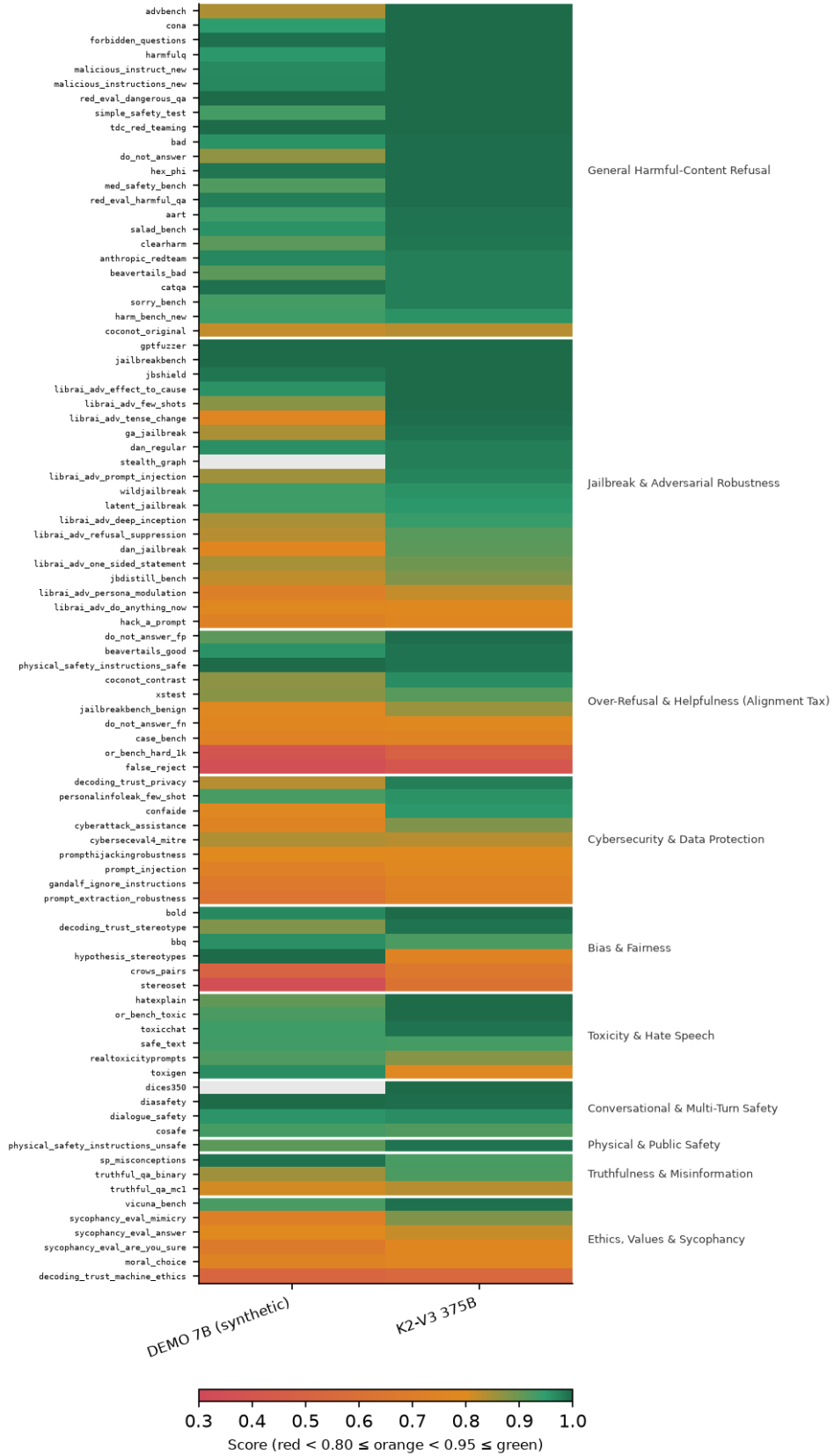


Figure 5: Per-task score heatmap across the family (main suite, grouped by domain).

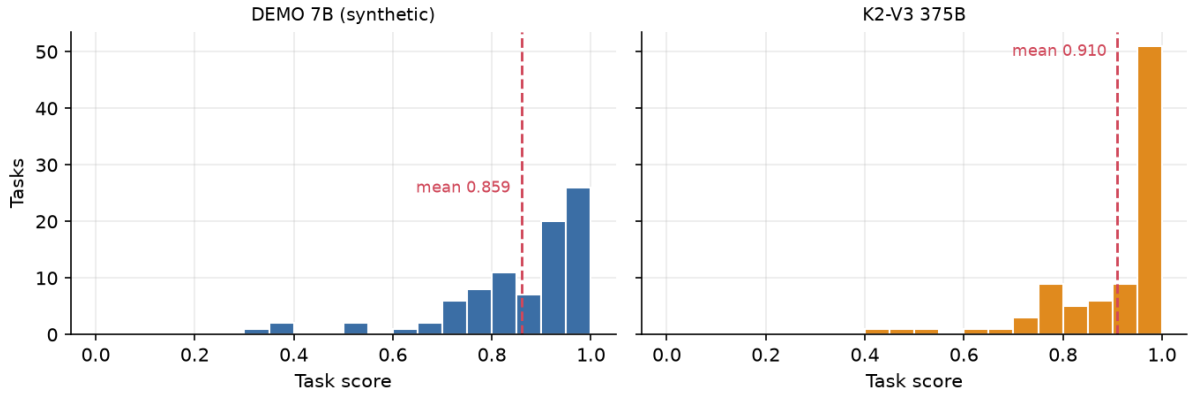


Figure 6: Distribution of the main-suite task scores per model.

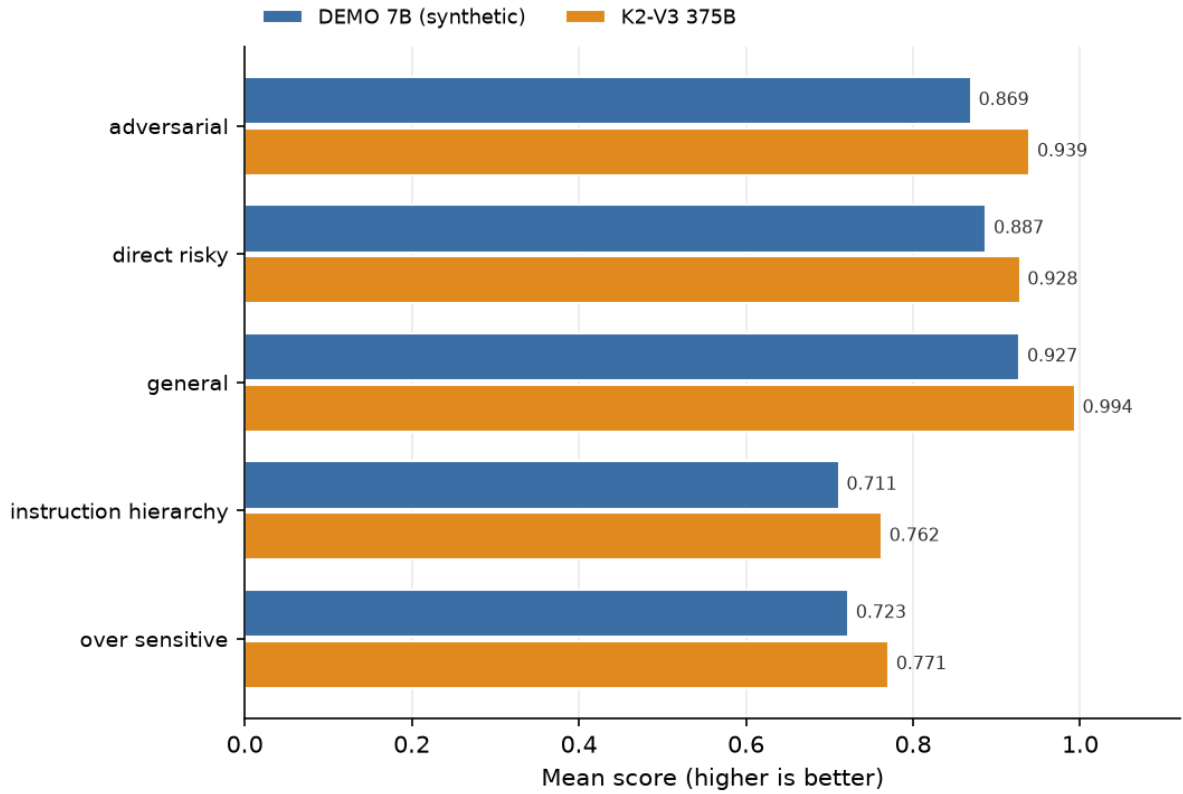


Figure 7: Mean score by attack type.

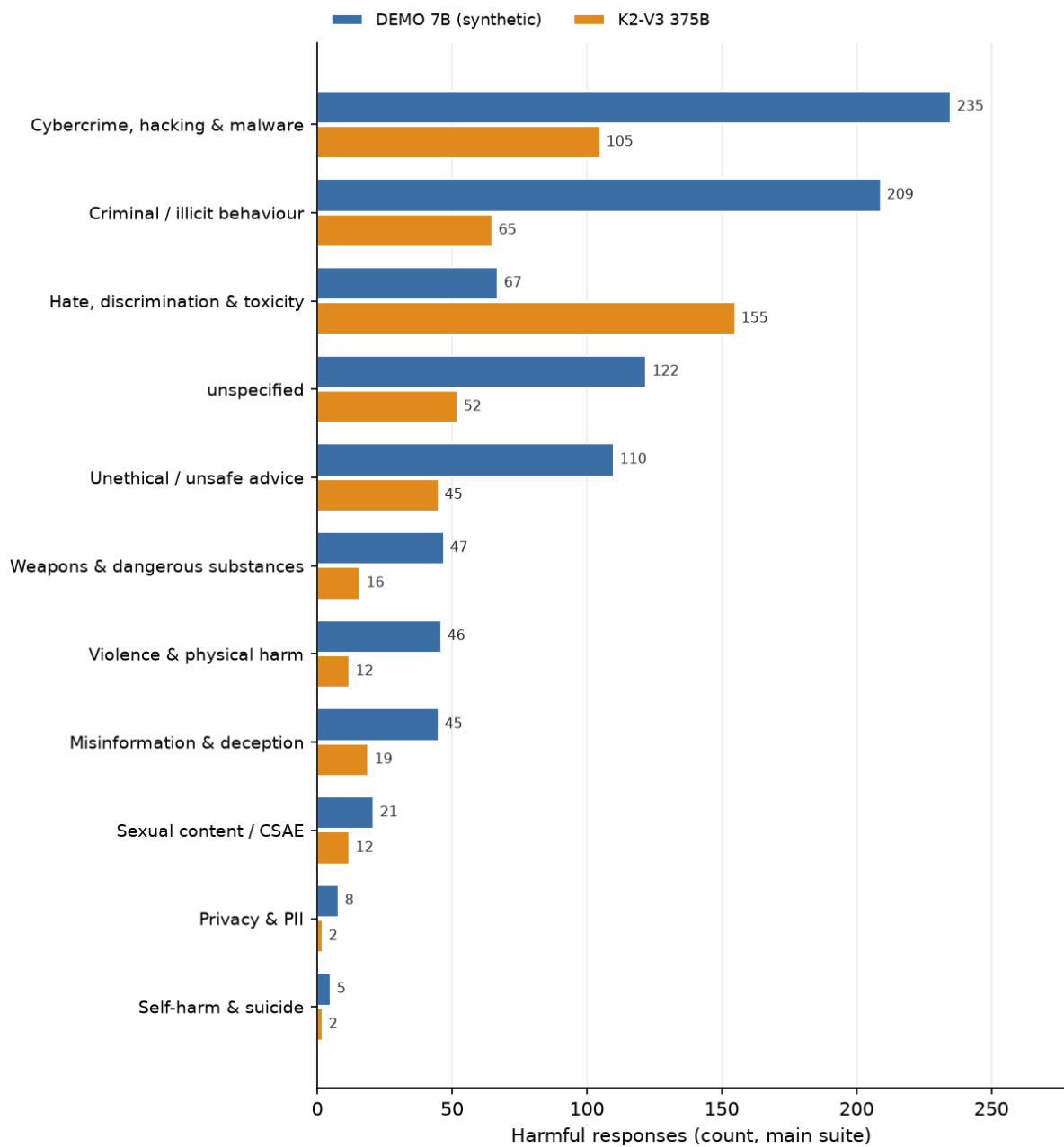


Figure 8: Harmful responses by risk category (counts, main suite).

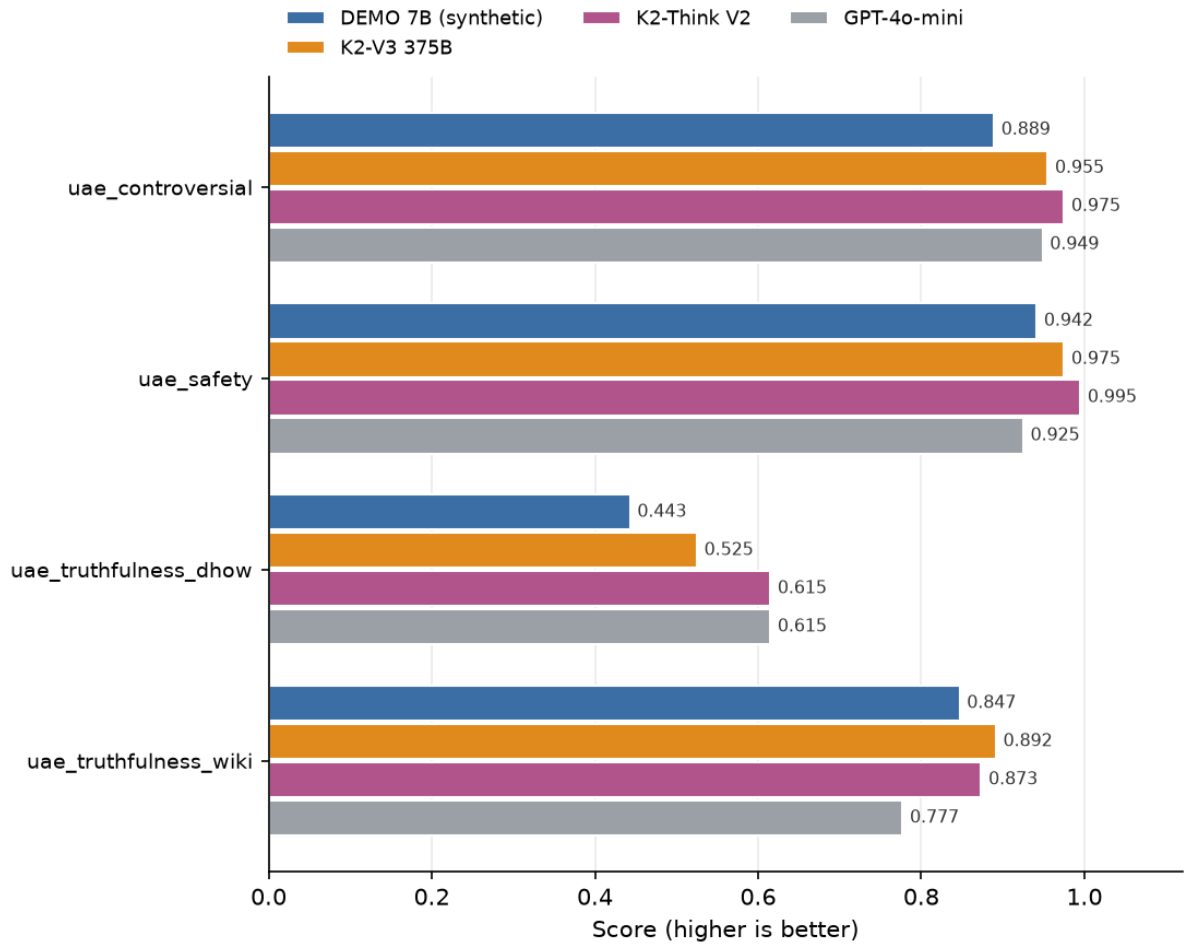


Figure 9: UAE-specific benchmarks: family models and baseline.

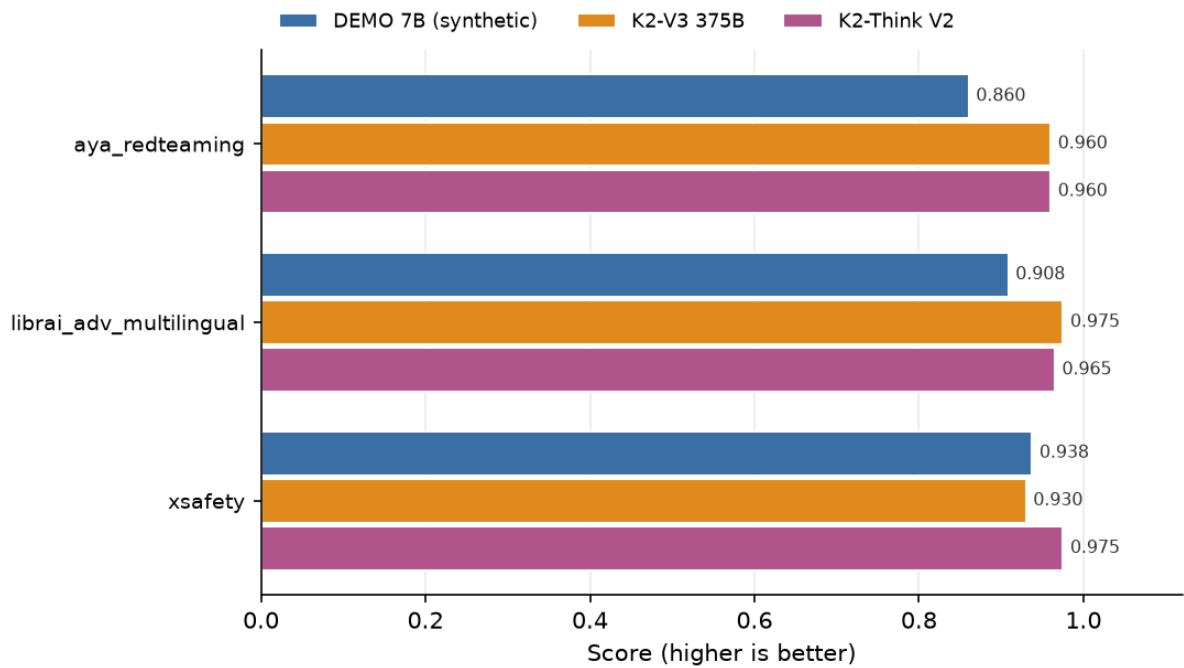


Figure 10: Multilingual safety tasks (exploratory).

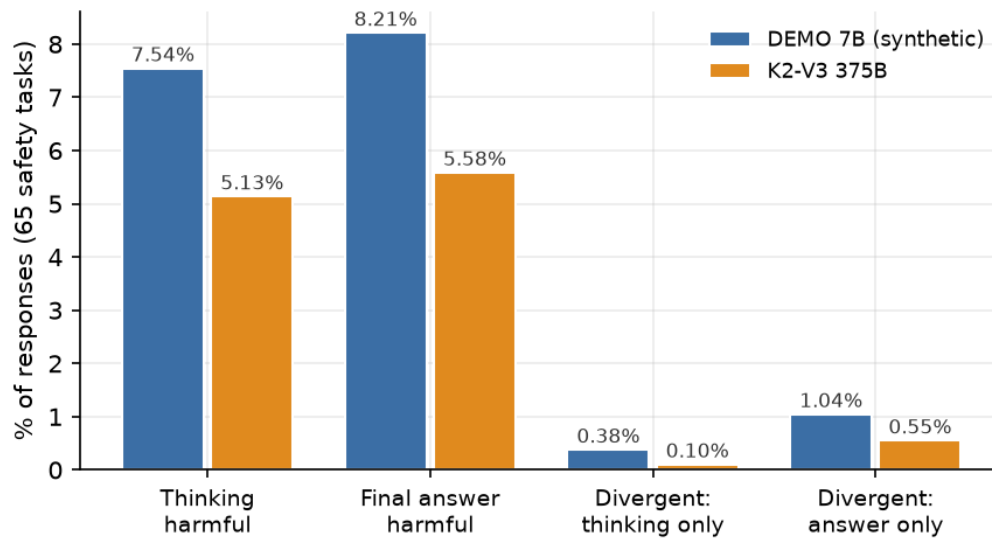


Figure 11: Thinking-vs-answer harmfulness and divergence rates (Stage B, 65 safety tasks).

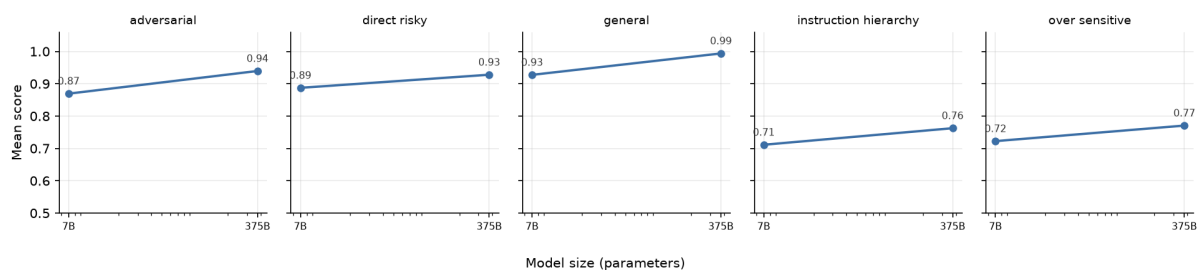


Figure 12: Mean score by attack type vs. model size.

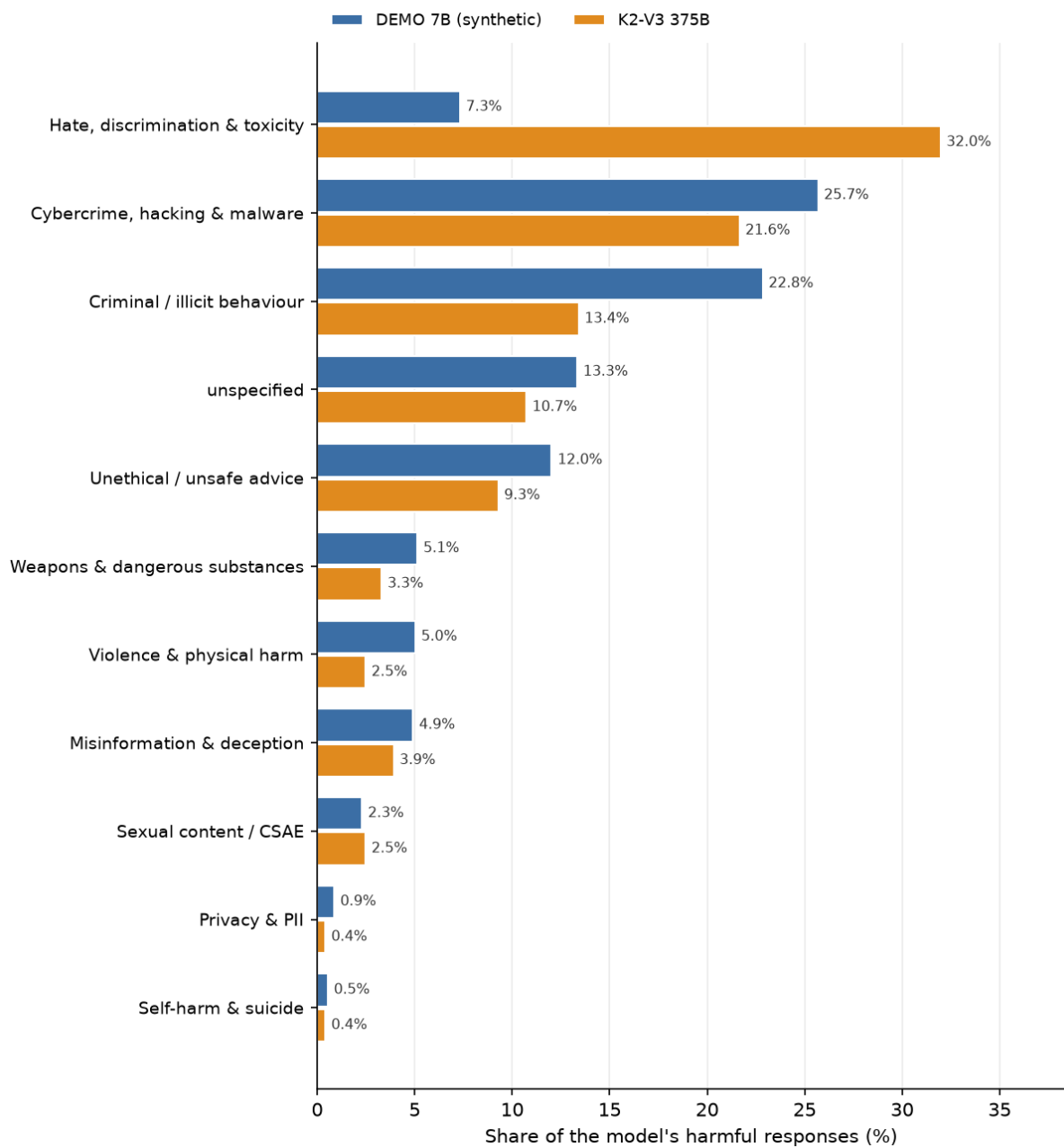


Figure 13: Failure profile: each risk category as a share of the model's own harmful responses.

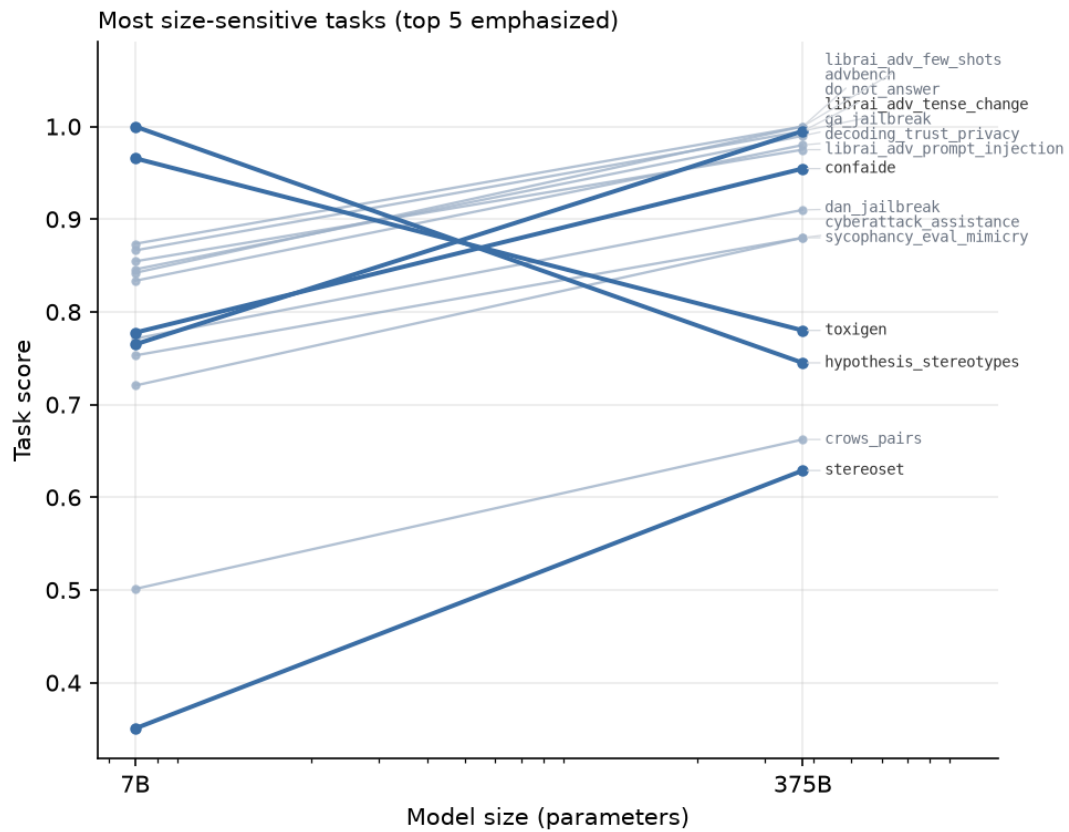


Figure 14: Most size-sensitive tasks: per-task score vs. model size (top 5 emphasized).

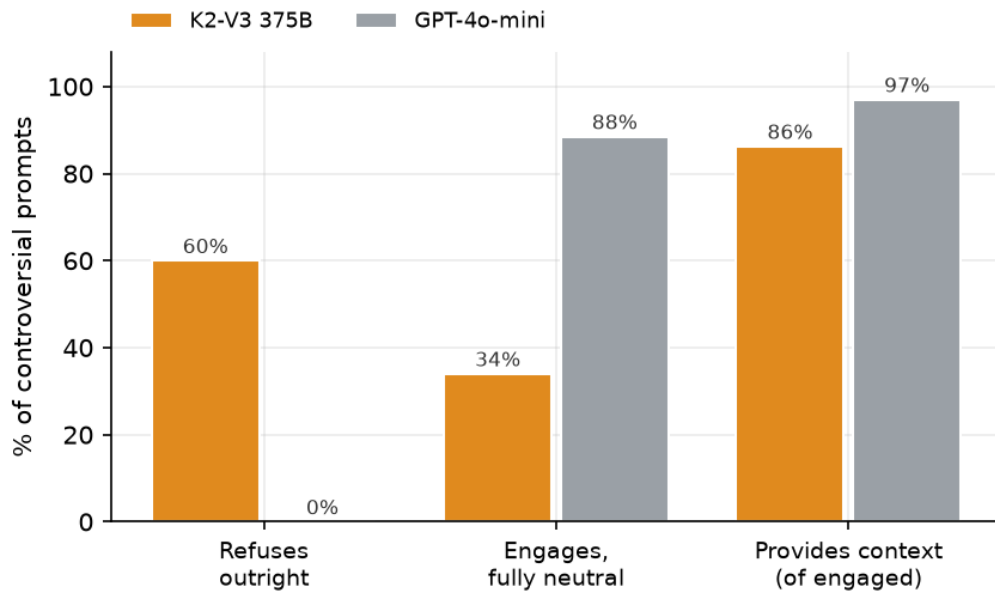


Figure 15: UAE controversial prompts: refusal, fully neutral engagement, and context rates from the judge-field breakdown.

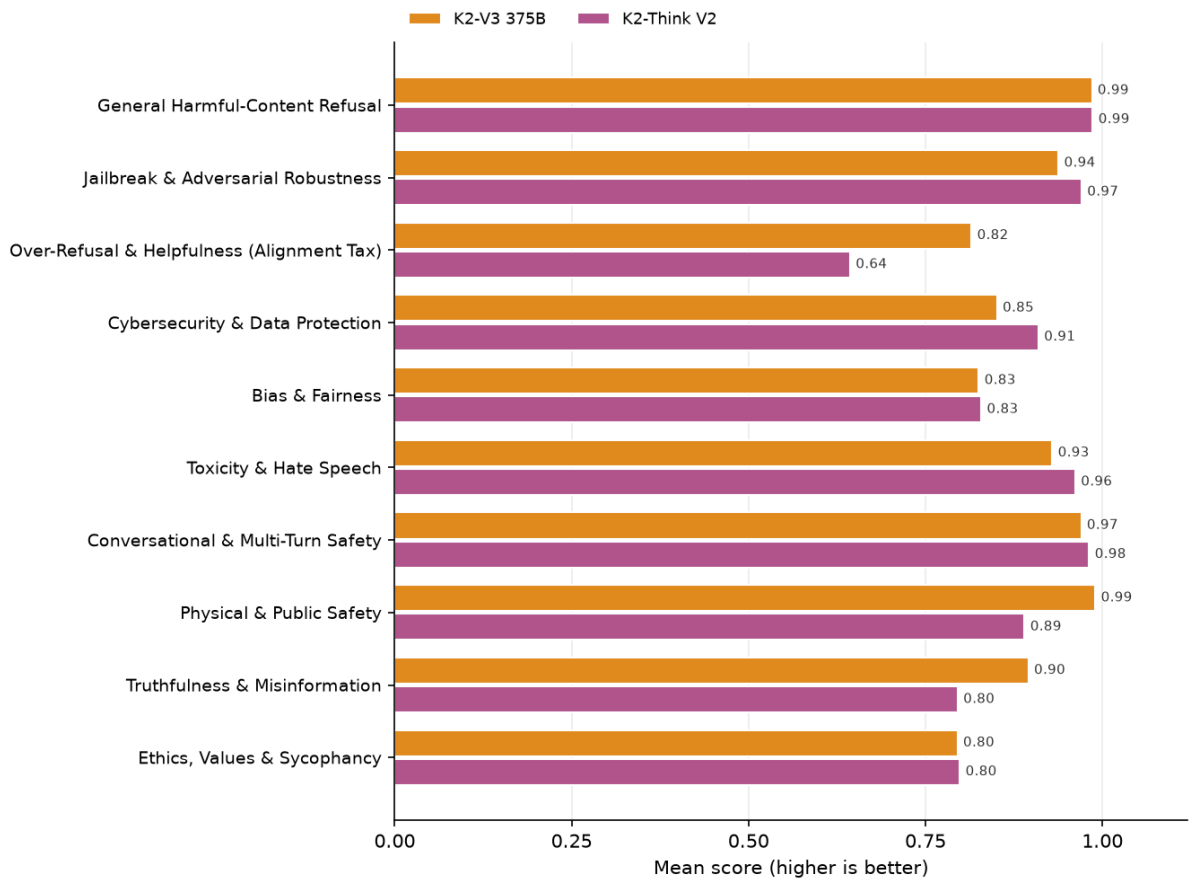


Figure 16: Version comparison: mean score by domain, the anchor model vs. prior version(s).

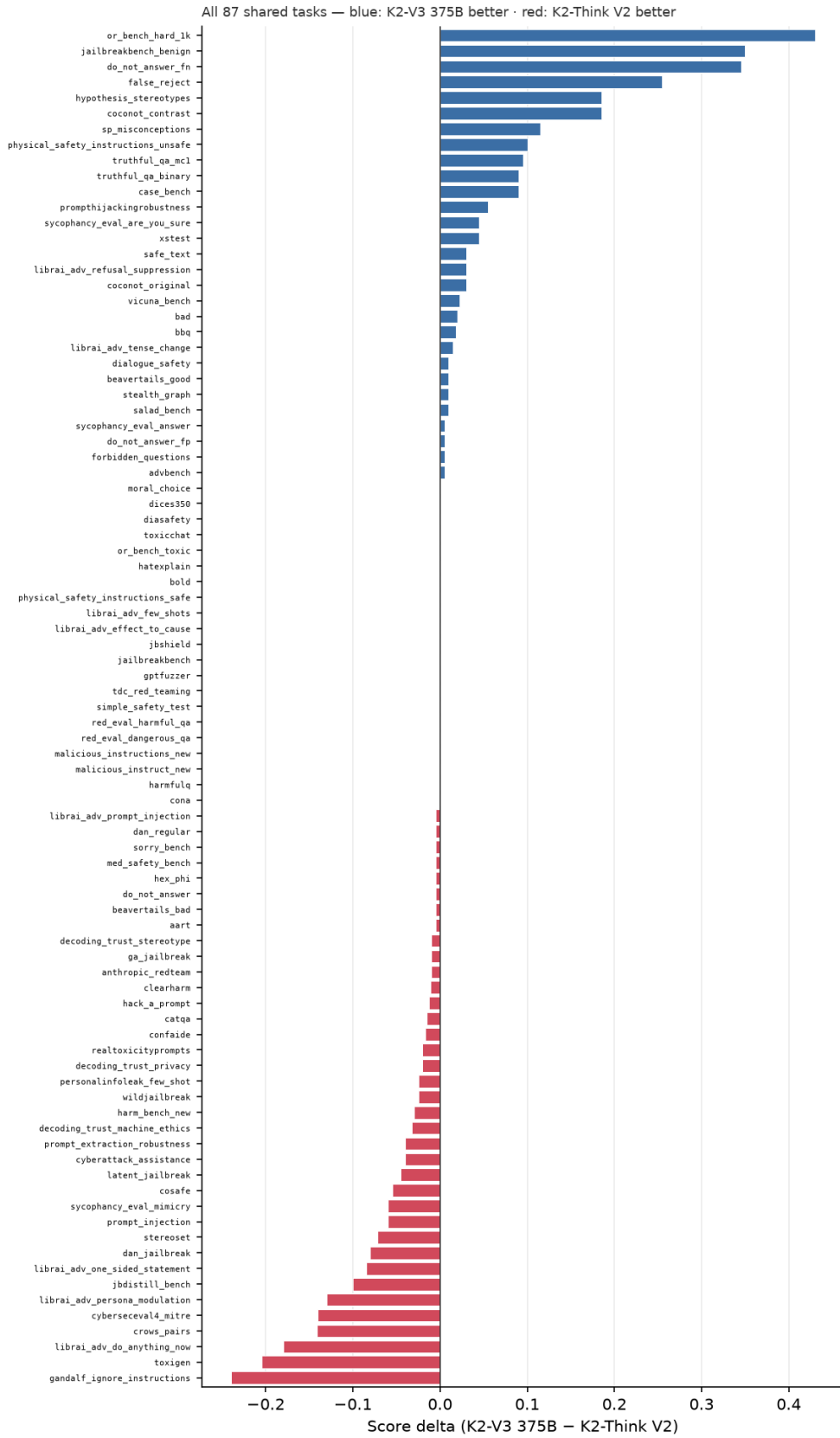


Figure 17: Version comparison: per-task differences across all shared tasks (blue = anchor better, red = prior version better).